

```
#include "FibonacciSequenceIterator.h"
```

```
FibonacciSequenceIterator::FibonacciSequenceIterator(  
    FibonacciSequence* aSequence, uint64_t aStart) noexcept  
    : fSequence(aSequence), fIndex(aStart) {}
```

```
const uint64_t& FibonacciSequenceIterator::operator*() const noexcept {  
    return **fSequence;  
}
```

```
FibonacciSequenceIterator& FibonacciSequenceIterator::operator++() noexcept ↗  
{  
    ++(*fSequence); // advance the sequence  
    fIndex++;       // advance the position  
    return *this;  
}
```

```
FibonacciSequenceIterator FibonacciSequenceIterator::operator++(int) ↗  
noexcept {  
    FibonacciSequenceIterator temp = *this;  
    ++(*this);  
    return temp;  
}
```

```
FibonacciSequenceIterator FibonacciSequenceIterator::begin() const noexcept ↗  
{  
    fSequence->begin(); // reset the underlying sequence  
    return FibonacciSequenceIterator(fSequence, 1); // start at position 1  
}
```

```
FibonacciSequenceIterator FibonacciSequenceIterator::end() const noexcept {  
    return FibonacciSequenceIterator(fSequence, MAX_FIBONACCI + 1); // stop ↗  
    at 94  
}
```

```
bool FibonacciSequenceIterator::operator==(  
    const FibonacciSequenceIterator& aOther) const noexcept {  
    return fIndex == aOther.fIndex;  
}
```